

Deep Learning for Time Series Analysis with Application on Human Rehabilitation

Journées Time series analysis and deep learning

Ali Ismail-Fawaz

MSD, IRIMAS, Université de Haute-Alsace, Mulhouse France

08 Juin 2026



1. Deep Learning for Time Series Data

2. Reducing Model Complexity

3. Imperfect Data

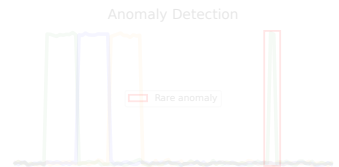
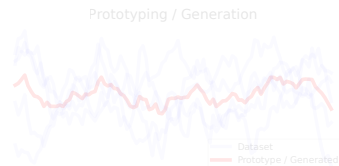
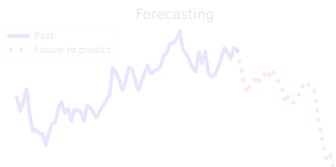
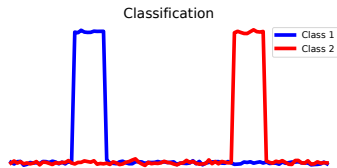
4. Application on Human Rehabilitation

5. Conclusion

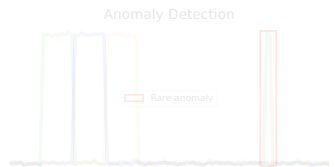
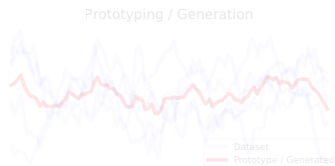
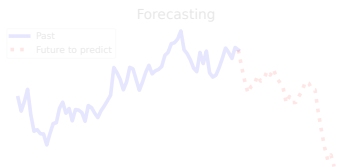
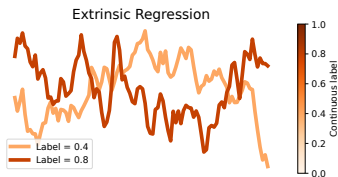
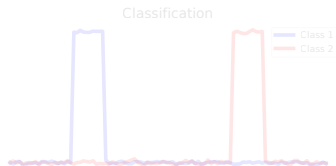
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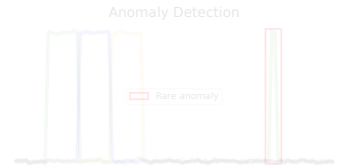
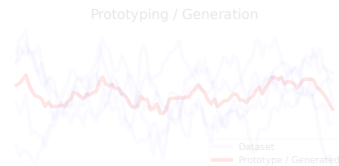
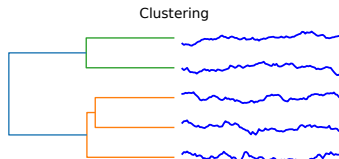
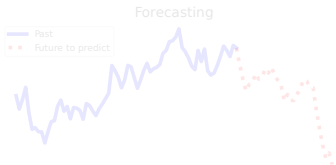
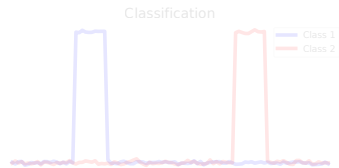
Time Series Machine Learning



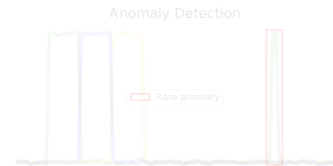
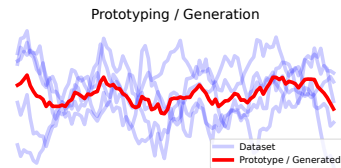
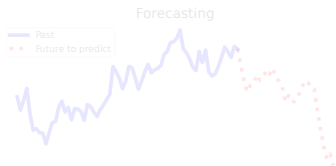
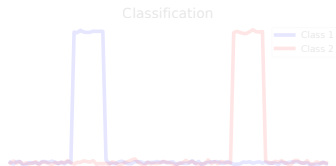
Time Series Machine Learning



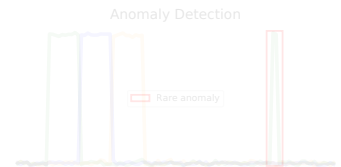
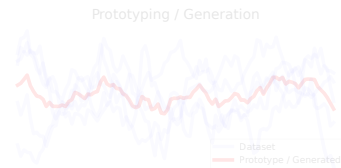
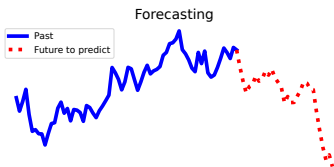
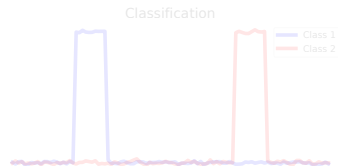
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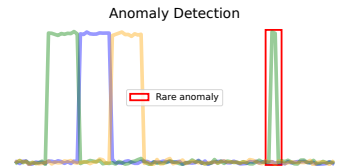
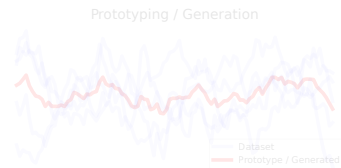
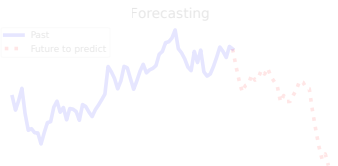
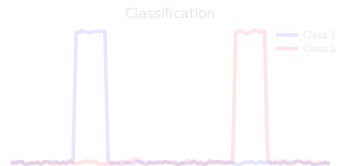
Time Series Machine Learning



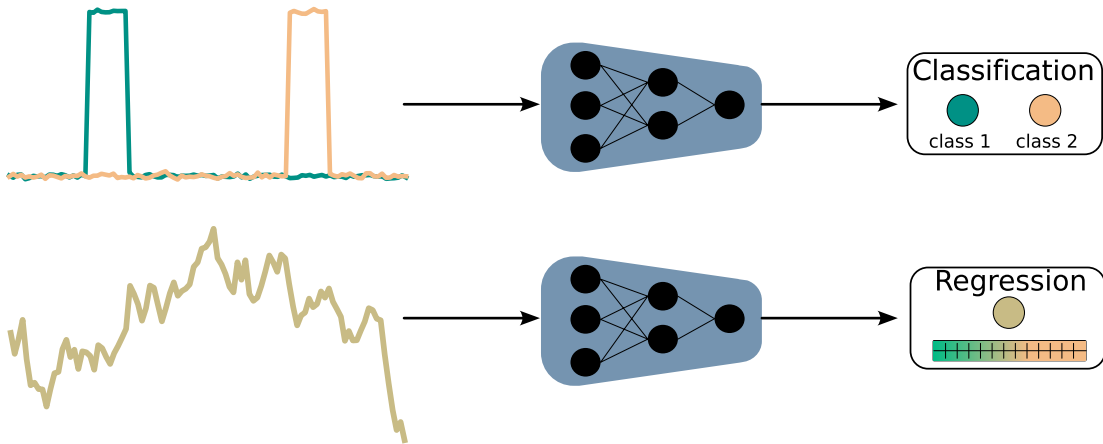
Time Series Machine Learning



Time Series Machine Learning

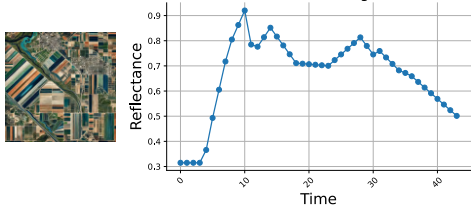


Deep Learning for Time Series Classification and Extrinsic Regression

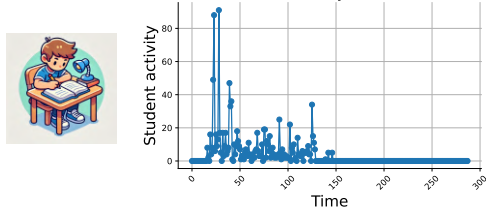


Time Series Classification / Extrinsic Regression Applications

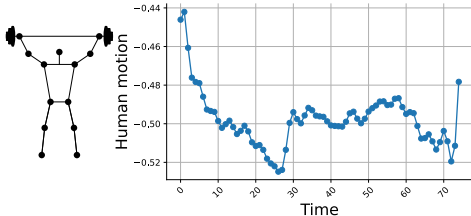
Time Series From Series of Satellite Images



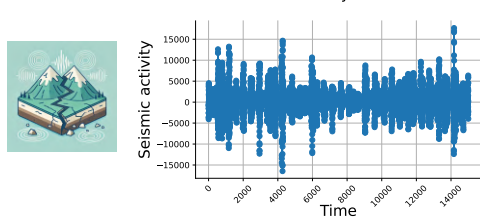
Time Series from Student activity



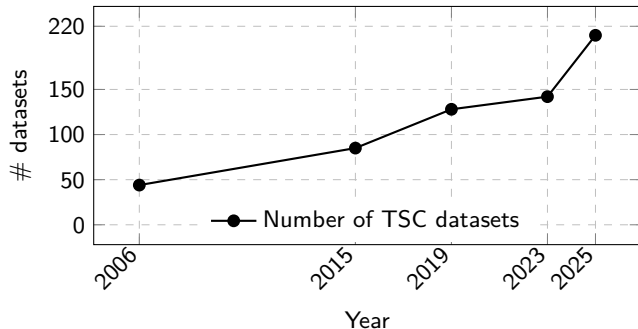
Time Series from Human motion



Time Series from Seismic activity



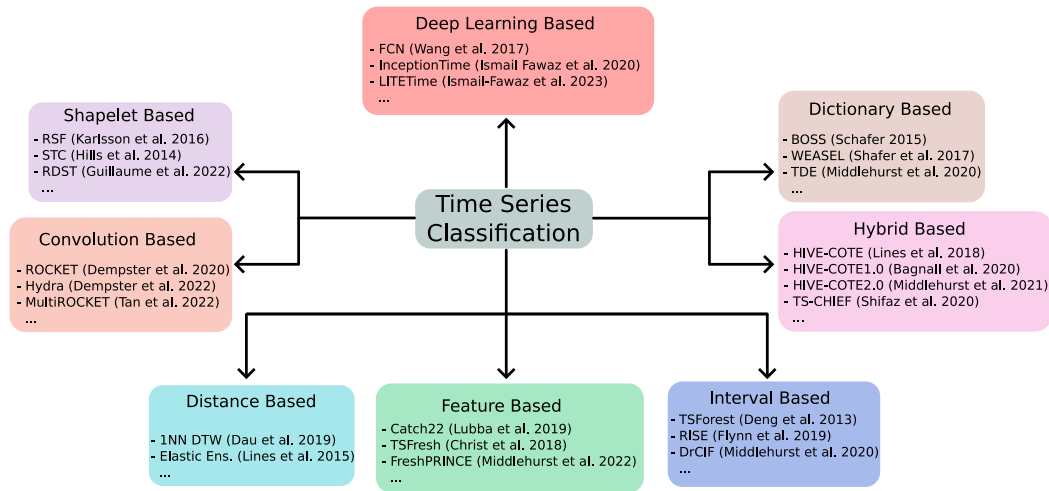
Evolution of the Time Series Classification (TSC) Archive



- 2006: UCR archive as a small but influential benchmark (44 datasets).
- 2015: Major expansion (85 datasets) $\Rightarrow$ broader evaluation for classical Machine Learning methods.
- 2019: Further growth (128 datasets) $\Rightarrow$ early deep learning models for TSC.
- 2023: 144 datasets $\Rightarrow$ richer diversity in domains, lengths, and difficulty.
- 2025: 210 datasets $\Rightarrow$ 29 and 39 new data from the MONSTER and Rehab-Pile archives

- Dau, H.A., Bagnall, A., Kamgar, K., Yeh, C.C.M., Zhu, Y., Gharghabi, S., Ratanamahatana, C.A. and Keogh, E., 2019. The UCR time series archive. *IEEE/CAA Journal of Automatica Sinica*, 6(6), pp.1293-1305.
- Middlehurst, M., Schäfer, P., & Bagnall, A. (2024). Bake off redux: a review and experimental evaluation of recent time series classification algorithms. *Data Mining and Knowledge Discovery*, 1-74.
- Dempster, A., Foumani, N.M., Tan, C.W., Miller, L., Mishra, A., Salehi, M., Pelletier, C., Schmidt, D.F. and Webb, G.I., 2025. MONSTER: Monash Scalable Time Series Evaluation Repository. *Journal of Data-centric Machine Learning Research*.
- Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J. and Forestier, G., A Standardized Benchmark for Skeleton-Based Rehabilitation Assessment Using Deep Learning. *IEEE International Conference on Face and Gestures*, 2026

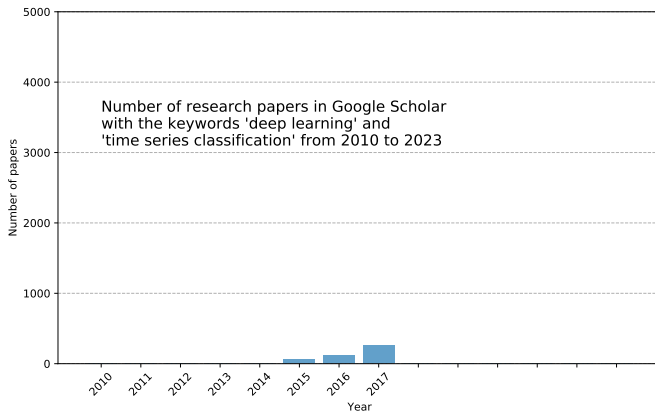
Taxonomy of methods



Middlehurst, M., Schäfer, P., & Bagnall, A. (2024). Bake off redux: a review and experimental evaluation of recent time series classification algorithms. *Data Mining and Knowledge Discovery*, 1-74.

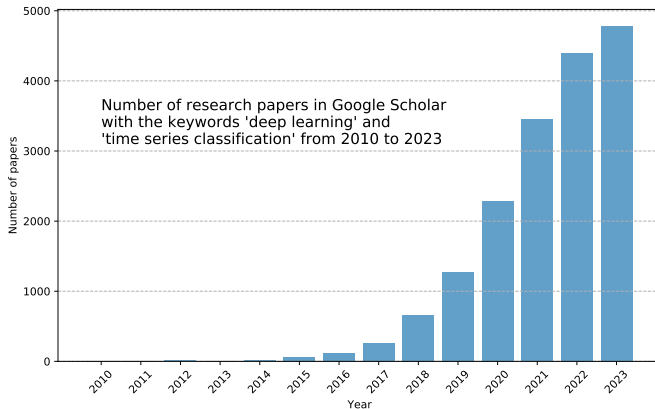
Why Deep Learning for TSC ?

- In 2017, reviewers started wondering what would be the performance of deep learning for TSC when reviewing non-deep learning TSC papers
- Deep learning had great success on other type of data, so why not on time series ?
- Deep Learning models learn to extract features and perform classification with one set of parameters



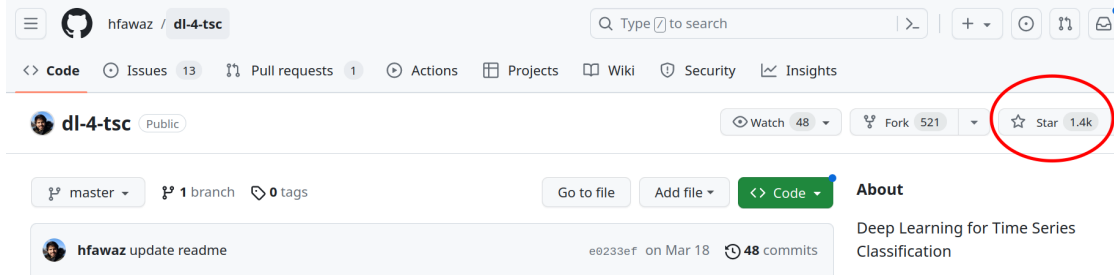
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In **2019**, we presented a study of **Deep Learning for Time Series Classification** (cited more than 4.8K times (GoogleScholar)) [1].

- We selected models with enough details (or available code) to reproduce the model's architecture
- We benchmarked all the models on the UCR archive [2]
- We published the code on Github for reproducibility and got very positive feedback (>1.4K stars)



hfawaz / dl-4-tsc

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About

Deep Learning for Time Series Classification

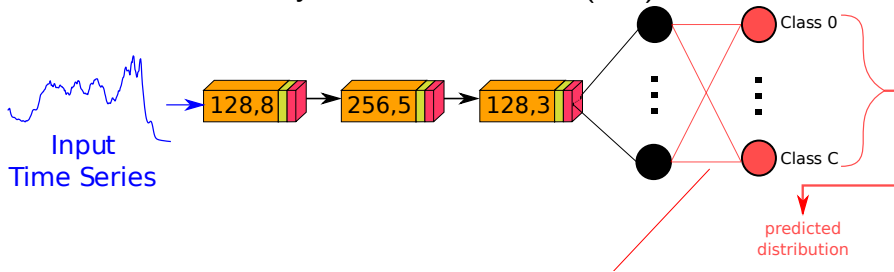
[1] Ismail Fawaz, H., Forestier, G., Weber, J., Idoumghar, L., & Muller, P. A. (2019). Deep learning for time series classification: a review. *Data mining and knowledge discovery*, 33(4), 917-963.

[2] Dau, H. A., Bagnall, A., Kamgar, K., Yeh, C. C. M., Zhu, Y., Gharghabi, S. & Keogh, E. (2019). The UCR time series archive. *IEEE/CAA Journal of Automatica Sinica*

 : 1D convolution layer with n filters of size k .  : batch normalization.  : activation

 : fully Connected.  : 1D global average pooling

Fully Convolutional Networks (FCN)



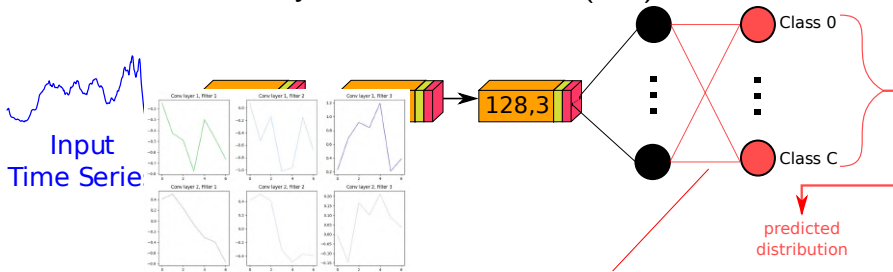
Wang, Z., Yan, W. and Oates, T., 2017, May. Time series classification from scratch with deep neural networks: A strong baseline. In 2017 International joint conference on neural networks (IJCNN) (pp. 1578-1585). IEEE.

DL4TSC - Some Architectures

 : 1D convolution layer with n filters of size k .  : batch normalization.  : activation

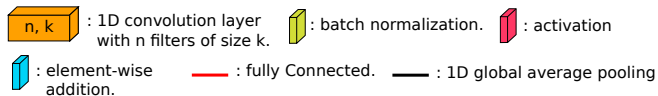
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Fully Convolutional Networks (FCN)

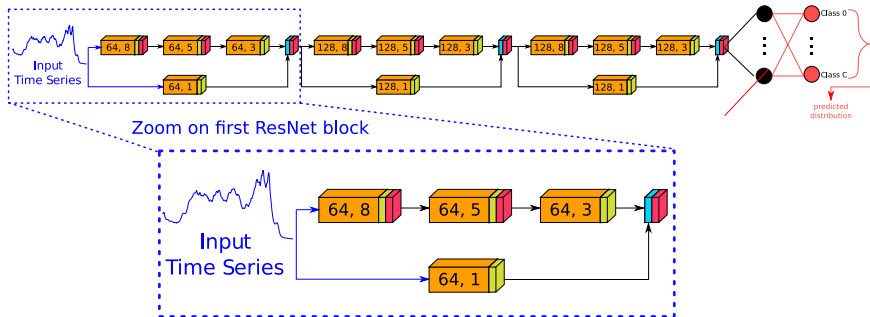


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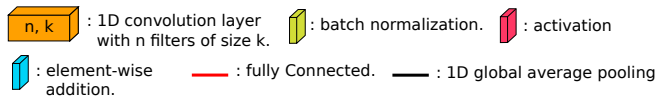
DL4TSC - Some Architectures



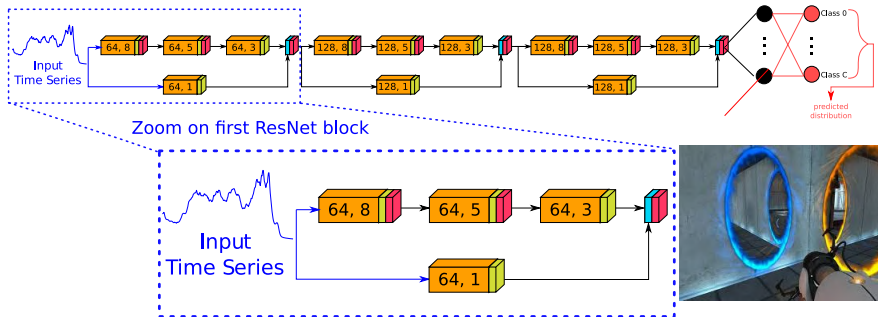
Residual Network (ResNet):



DL4TSC - Some Architectures

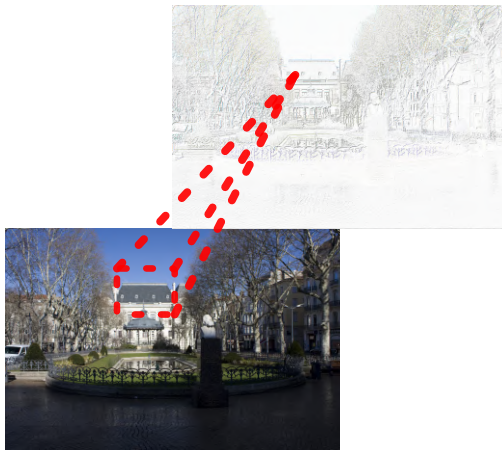


Residual Network (ResNet):



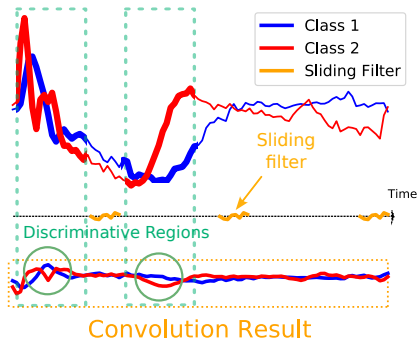
DL4TSC - Some Architectures

Convolutions on Images vs Time Series



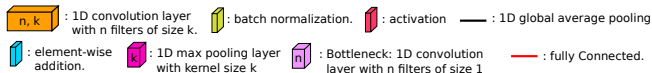
The result of applying an edge detection convolution on an image

Image source: Daniel VILLAFRUELA

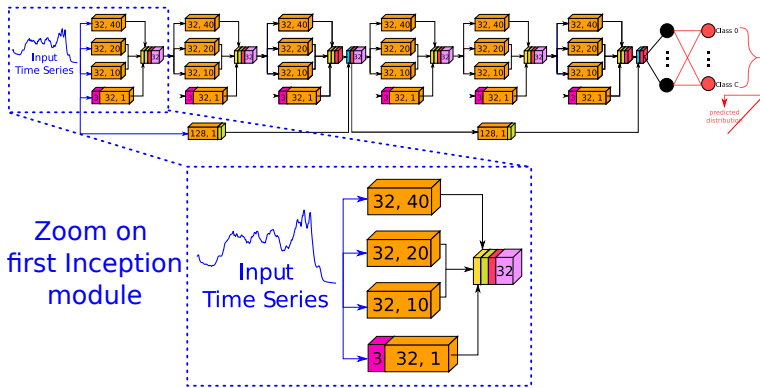


The result of applying a learned discriminative convolution on the ECG200 dataset

DL4TSC - Some Architectures



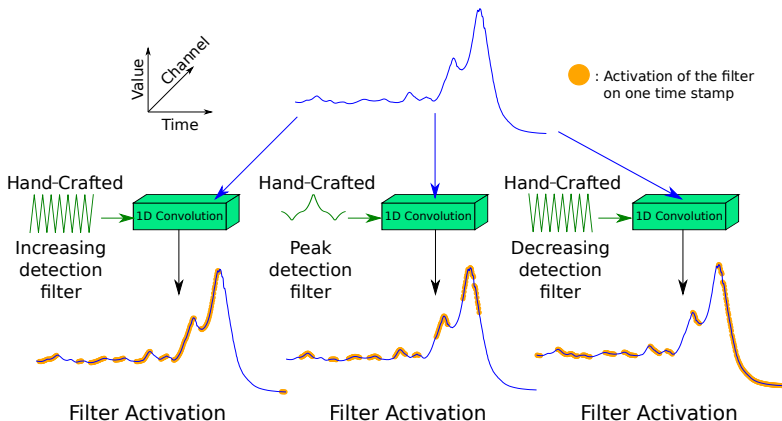
Inception



Ismail Fawaz, H., Lucas, B., Forestier, G., Pelletier, C., Schmidt, D.F., Weber, J., Webb, G.I., Idoumghar, L., Muller, P.A. and Petitjean, F., 2020. Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery*, 34(6), pp.1936-1962.

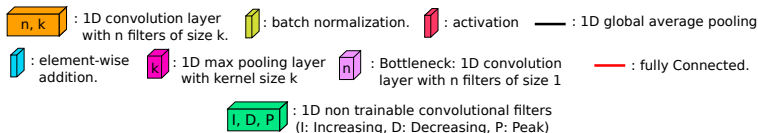
Hand-Crafted Filters

Bridging the Gap Between Random and Fully Learned Filters

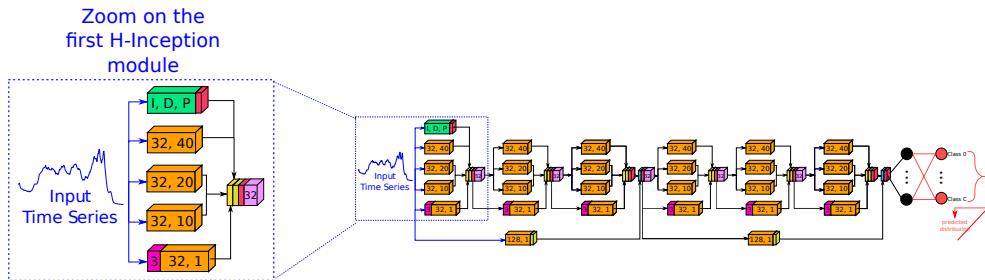


Ismail-Fawaz, A., Devanne, M., Weber, J., & Forestier, G. (2022). Deep learning for time series classification using new hand-crafted convolution filters. In 2022 IEEE International Conference on Big Data (Big Data) (pp. 972-981)

Hybrid Inception



Hybrid Inception (H-Inception) Architecture



Ismail-Fawaz, A., Devanne, M., Weber, J., & Forestier, G. (2022). Deep learning for time series classification using new hand-crafted convolution filters. In 2022 IEEE International Conference on Big Data (Big Data) (pp. 972-981)

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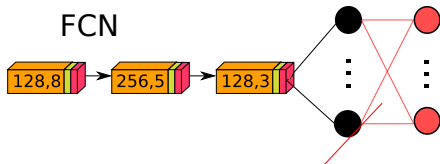
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Reducing Model Complexity

Deep learning models can be difficult to deploy:

Deep learning model
with high number
of parameters

Example: robotic systems
for ground type detection



Deployment



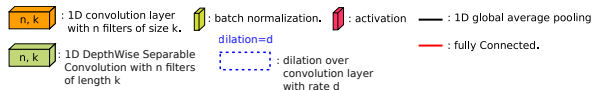
© <http://timeseriesclassification.com/>

I don't have enough
memory and resources
to handle such complex
model !

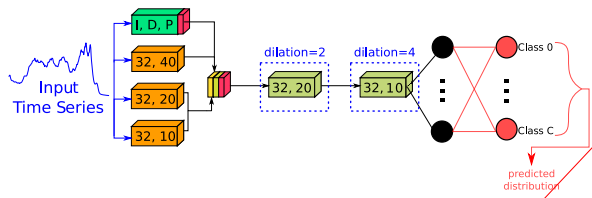
- Is it possible to reduce complexity without compromising performance?

LITE: Light Inception with boostTing tEchniques

- Replace standard convolution with depthwise separable convolution to reduce the number of parameters
- Compensate the loss of complexity by:
 1. Use hand-crafted convolutional filters
 2. Use dilated convolution
 3. Use multiplexing convolution



LITE Architecture



■ **Ismail-Fawaz, A.**, Devanne, M., Berretti, S., Weber, J. and Forestier, G., 2023, October. Lite: Light inception with boosting techniques for time series classification. In 2023 **IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA)** (pp. 1-10). IEEE.

Reducing Models Complexity: LITE - Efficiency Comparison

Table: Comparison between the proposed method with FCN, ResNet and Inception without ensemble.

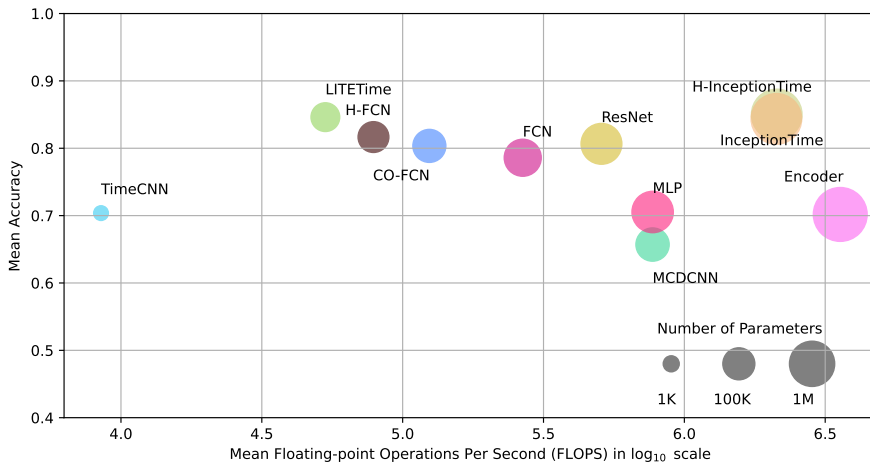
Models	Number of parameters	FLOPS	Training Time (mins)	Testing Time (mins)	CO2 (g)
Inception	420,192	424,414	2,419.2	1.296	0.2928
ResNet	504,000	507,818	2,750.4	1.008	0.3101
FCN	264,704	266,850	2,491.2	0.4464	0.2623
LITE	9,814	10,632	829.8	0.72	0.1048

CO2 calculation: <https://codecarbon.io/>

- **Ismail-Fawaz, A.**, Devanne, M., Berretti, S., Weber, J. and Forestier, G., 2023, October. Lite: Light inception with boosting techniques for time series classification. In 2023 **IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA)** (pp. 1-10). IEEE.

Comparing all the architectures

We created a dynamic website (updated regularly) to compare all these architectures in terms of performance and complexity on time series classification data:



128 datasets Try it out on : <https://msd-irimas.github.io/pages/dl4tsc/>

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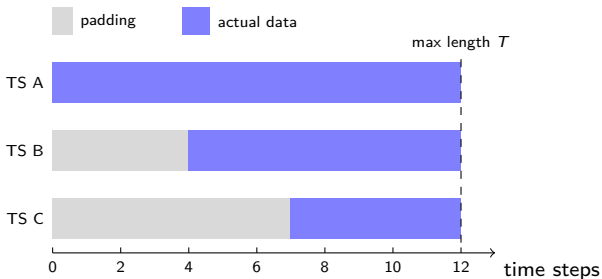
Imperfect Data in Deep Learning for Time Series Classification

Unequal Length:

Deep models require all time series in a batch to have the same length.

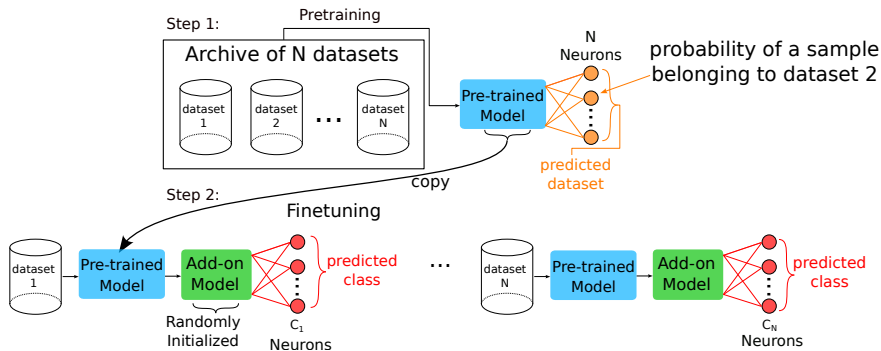
Two main strategies:

1. Global padding: pad all series (from the beginning) with zeros up to the maximum length.
2. Batch-wise padding: pad only to the longest time series within each batch.



Unequal Length: Use Case - Foundation Models

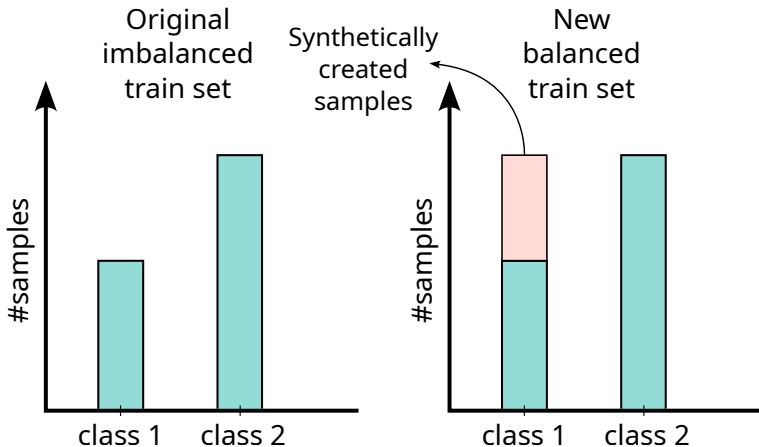
1. Instead of training on a single large or small dataset, train on multiple datasets at the same time (unsupervised pretext task)
2. Fine-tune the pre-trained model on the classification task of each dataset alone
3. Pretext task: predicting the original dataset of each time series



Imperfect Data in Deep Learning for Time Series Classification

Class Imbalance or Not Enough Data:

Handled via data augmentation or re-sampling.

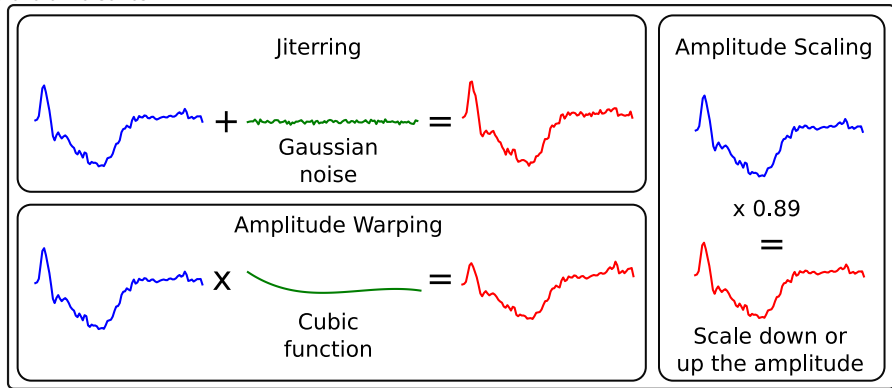


The supervised loss can be weighted during training based on the number of samples per class in each batch.

Data Augmentation

How to create synthetic time series ?

- We add noise to the series
- We scale the time series

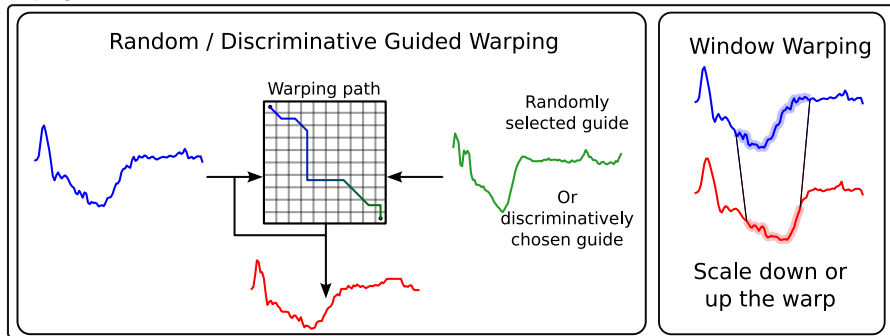


- Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J. and Forestier, G., Re-framing Time Series Augmentation Through the Lens of Generative Models. *ECML/PKDD Workshop on Advanced Analytics and Learning on Temporal Data*, 2025
- Um, T.T., Pfister, F.M., Pichler, D., Endo, S., Lang, M., Hirche, S., Fietzek, U. and Kulić, D., 2017, November. Data augmentation of wearable sensor data for parkinson's disease monitoring using convolutional neural networks. In *Proceedings of the 19th ACM international conference on multimodal interaction* (pp. 216-220).

Data Augmentation

How to create synthetic time series ?

- We warp a time series using the warping path with another series
- We add warping distortions to a window in the time series

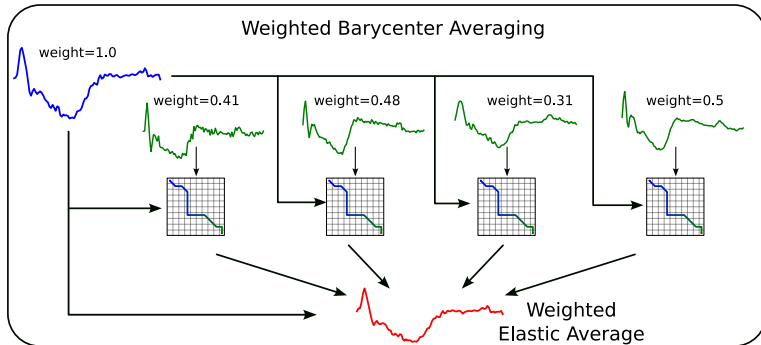


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- Iwana, B.K. and Uchida, S., 2021, January. Time series data augmentation for neural networks by time warping with a discriminative teacher. In *2020 25th International Conference on Pattern Recognition (ICPR)* (pp. 3558-3565). IEEE.
- Le Guennec, A., Malinowski, S. and Tavenard, R., 2016, September. Data augmentation for time series classification using convolutional neural networks. In *ECML/PKDD workshop on advanced analytics and learning on temporal data*.

Data Augmentation

How to create synthetic time series ?

- We averaged a set of time series and took the average as a new synthetic object
- We used weighted averages to generate multiple synthetic objects



- Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J. and Forestier, G., Re-framing Time Series Augmentation Through the Lens of Generative Models. *ECML/PKDD Workshop on Advanced Analytics and Learning on Temporal Data*, 2025
- Petitjean, F., Ketterlin, A., & Gançarski, P. (2011). A global averaging method for dynamic time warping, with applications to clustering. *Pattern Recognition*, 44(3), 678-693.
- Forestier, G., et al. "Generating synthetic time series to augment sparse datasets." 2017 *IEEE International Conference on Data Mining (ICDM)*. IEEE, 2017.
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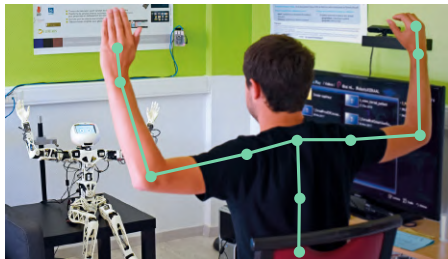
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1. Deep Learning for Time Series Data
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- Rehabilitation assessment is critical for:
 - Monitoring patient recovery
 - Evaluating exercise quality
 - Supporting personalized therapy

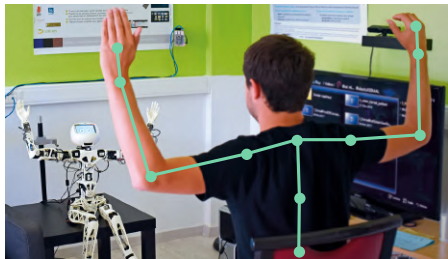
Context and Motivation

- Rehabilitation assessment is critical for:
 - Monitoring patient recovery
 - Evaluating exercise quality
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- Video-based skeleton extraction enables:
 - Low-cost assessment
 - Remote rehabilitation
 - Smartphone/webcam deployment



Context and Motivation

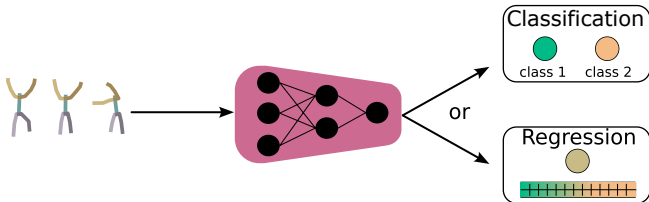
- Rehabilitation assessment is critical for:
 - Monitoring patient recovery
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- Video-based skeleton extraction enables:
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 - Remote rehabilitation
 - Smartphone/webcam deployment
- Deep learning methods are increasingly used for skeleton-based rehabilitation assessment



Problem Statement

Rehabilitation Assessment

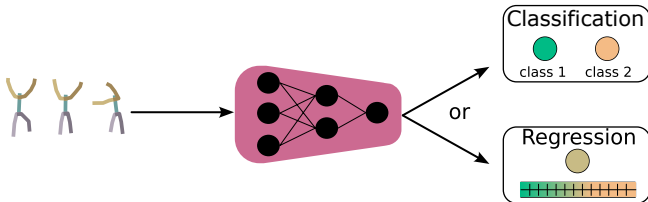
- Evaluates movement quality
- Classification example:
 - Correct vs incorrect squat
- Extrinsic regression example:
 - Score of correctness



Problem Statement

Rehabilitation Assessment

- Evaluates movement quality
- Classification example:
 - Correct vs incorrect squat
- Extrinsic regression example:
 - Score of correctness



Main Issues in the Literature

- Evaluations are performed on very few datasets
 - Mainly KIMORE and UI-PRMD
- Inconsistent experimental protocols
- Limited reproducibility and unavailable code

The Rehab-Pile

- We aggregated 8 existing repositories:

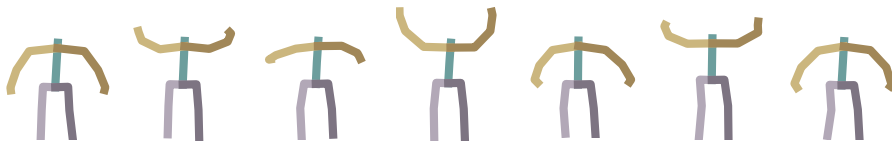
EHE
IRDS

KERAAL
SPHERE

KIMORE
UCDHE

KINECAL
UI-PRMD

- Each exercise is treated as a separate dataset



Category	# Datasets	# Sequences
Classification	39	14 721
Extrinsic regression	21	2 550
Total	60	17 291

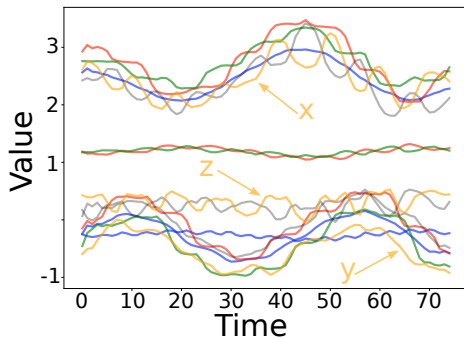
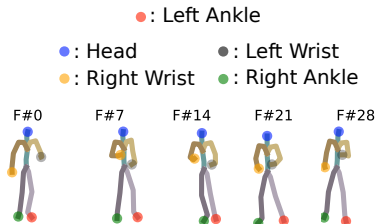


Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J. and Forestier, G., A Standardized Benchmark for Skeleton-Based Rehabilitation Assessment Using Deep Learning. [IEEE International Conference on Face and Gestures](#), 2026

Skeleton Sequence Representation

- Input: sequence of human skeletons extracted from video
- Each skeleton contains J joints in D dimensions (2D, 3D or more)
- A sequence is represented as a multivariate time series of length L :

$$S \in \mathbb{R}^{L \times J \times D}$$



Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J. and Forestier, G., A Standardized Benchmark for Skeleton-Based Rehabilitation Assessment Using Deep Learning. *IEEE International Conference on Face and Gestures*, 2026

Diverse Deep Learning Architectures

- Various existing architectures proposed for time series or human motion



Convolution
H-Inception
LITEMV



Recurrent
MotionGru
ConvLSTM



Graph
STGCN



Transformer
Vanilla Transformer
ConvTran

- **Ensembles of 5 models randomly initialized**

Cross-subject evaluation

- Test sets contain **only unhealthy** subjects whenever possible
- 5-fold if # subjects ≥ 10
- Leave-One-Subject-Out if # subjects < 10



Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J. and Forestier, G., A Standardized Benchmark for Skeleton-Based Rehabilitation Assessment Using Deep Learning. *IEEE International Conference on Face and Gestures*, 2026

Classification and Regression Results

Model	Accuracy (%)	Balanced Accuracy (%)
STGCN	64.55 ± 14.91	57.65 ± 10.68
ConvLSTM	67.08 ± 16.37	62.59 ± 13.88
ConvTran	70.61 ± 12.90	65.72 ± 09.69
H-Inception	70.78 ± 12.75	65.67 ± 10.72
MotionGRU	72.82 ± 12.22	<u>67.58 ± 11.88</u>
VanTran	<u>74.30 ± 13.34</u>	67.47 ± 13.79
LITEMV	75.49 ± 11.28	70.34 ± 11.18

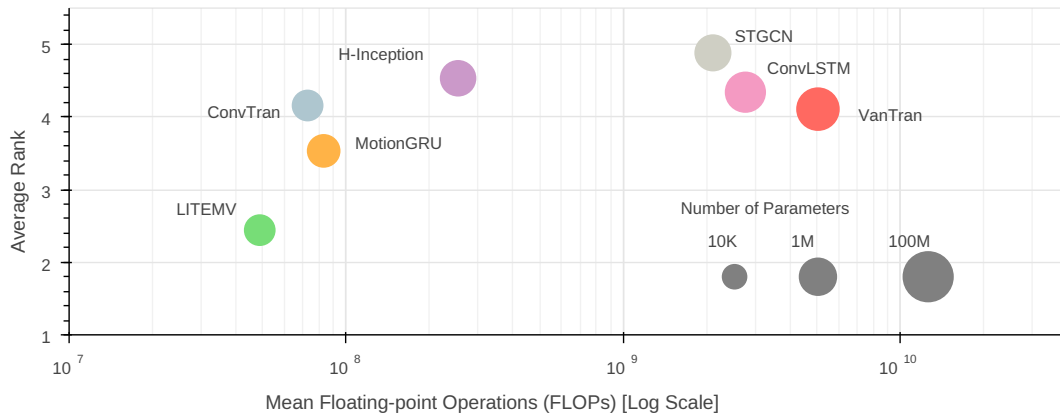
LITEMV, **VanTran** and **MotionGRU** are the most accurate

Classification and Regression Results

Model	RMSE	MAE
STGCN	8.1084 ± 12.1624	6.7389 ± 10.2665
ConvLSTM	7.9354 ± 11.9593	6.5461 ± 10.0711
ConvTran	<u>7.3491 ± 10.8790</u>	<u>6.0729 ± 09.0227</u>
H-Inception	7.9342 ± 11.7514	6.3744 ± 09.3453
MotionGRU	7.8016 ± 11.6802	6.2966 ± 09.5963
VanTran	8.1758 ± 12.5573	6.8688 ± 10.5411
LITEMV	6.4286 ± 09.6731	5.3497 ± 08.1396

LITEMV and **ConvTran** are the most accurate

Performance vs Efficiency Trade-off



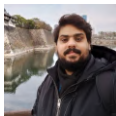
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Conclusion

The team



Ali
Ismail-Fawaz



Hassan
Ismail Fawaz



Maxime
Devanne



Stefano
Berretti



Jonathan
Weber



Germain
Forestier

Open-Source: GitHub: MSD-IRIMAS and aeon-toolkit

Hot Topics:

- Data Augmentation for Time Series Data
- Self-supervised Learning / Representation Learning
- Foundation Models for Time Series Data [1]
- Transformers for Time Series Classification [2]
- Multivariate Time Series Classification

■ [1] Ismail-Fawaz, A., Devanne, M., Berretti, S., Weber, J. and Forestier, G. "Finding foundation models for time series classification with a pretext task." *PAKDD International Workshop on Temporal Analytics*, 2024

■ [2] Foumani, N.M., Tan, C.W., Webb, G.I. and Salehi, M., 2024. Improving position encoding of transformers for multivariate time series classification. *Data mining and knowledge discovery*, 38(1), pp.22-48.